

Mileage Tracker v3 — Build Report

Story: Build Mileage Tracker v3 with polished UI, dark mode, and mobile responsive layout
Repository: raghu-cmyk/mileage-tracker-v3 | Branch: feat/mvp-v3
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1. Job Summary

Order	Job ID	Type	State	Duration
1	20260608T195322Z_b08e17	plan	completed	1m 3s
2	20260608T195548Z_30052b	implement	completed	2m 25s
3	20260608T195839Z_3093cd	validate	completed	54s
4	20260608T195952Z_526553	deliver	completed	1m 0s
5	20260608T200108Z_73108e	report	running	0s

2. Requirements → GitHub Issues Mapping

Issue	Title	Label	Requirements Covered
#1	Design & Aesthetic — Mileage Tracker v3 Design System	design-system	\$8.1 FR-UI1; Ledger palette, typography, layout, components
#2	Authentication & Secure Session Management	plan-task	\$5.1 FR-A1–A4; Argon2id, sessions, rate limiting
#3	Vehicle Management with Annual Odometer Readings	plan-task	\$5.2 FR-V1–V2; §274(d) odometer, archive/audit
#4	Trip Management with IRC §274(d) Substantiation	plan-task	\$5.3 FR-T1–T5; \$5.4 FR-C1–C2; audit trail
#5	IRS Mileage Rates & Trip Categories Reference Data	plan-task	\$4.4–4.5; 2025/2026 seeded rates, no hardcoding
#6	Deduction Calculation & Year-End Summary Report	plan-task	\$5.5 FR-R1–R3; business-use %, integer cents
#7	IRS-Compliant CSV & PDF Export	plan-task	\$5.6 FR-E1–E3; substantiation columns, accountant PDF
#8	Dark Mode with Persistent Theme Toggle	plan-task	\$8.2 FR-UI2; CSS custom properties, WCAG AA
#9	Mobile-Responsive Layout (360px–Desktop)	plan-task	\$8.3 FR-UI3; hamburger nav, reflowing tables
#10	Polished UI with Shared Design Tokens	plan-task	\$8.1 FR-UI1; cohesive tokens across all pages

Delivery PR: #11 — feat: Mileage Tracker v3 MVP (head: feat/mvp-v3 → base: mc_webhook_test)

3. Per-Job Agent Activity Logs

Job 20260608T195322Z_b08e17 (plan)

Planning complete. I read `docs/requirements/mileage-tracker-requirements.md` from `raghu-cmyk/mileage-tracker-v3` and created **10 GitHub issues** — 1 design system issue plus 9 implementation plan-task issues covering the full MVP and Section 8 enhancements.

```
## Design System

| # | Issue | Label |
|---|-----|-----|
| 1 | [Design & Aesthetic – Mileage Tracker v3 Design System](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/1) | `design` |

Defines the "Ledger" palette (light + dark hex tokens), Inter/JetBrains Mono typography, 12-column grid with breakpoints, and component specs (buttons, forms, tables, states, accessibility).
## MVP Features (`plan-task`)
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#	Issue	Covers
2	[Authentication & Secure Session Management](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/2)	FR-A1-A4, Argon2id, sessions, rate limiting
3	[Vehicle Management with Annual Odometer Readings](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/3)	FR-V1-V2, §274(d) odometer, archive/audit
4	[Trip Management with IRC §274(d) Substantiation](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/4)	FR-T1-T5, late-entry flagging, audit trail
5	[IRS Mileage Rates & Trip Categories Reference Data](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/5)	§4.4-4.5, 2025/2026 seeded rates, no hardcoding
6	[Deduction Calculation & Year-End Summary Report](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/6)	FR-R1-R3, business-use %, integer cents
7	[IRS-Compliant CSV & PDF Export](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/7)	FR-E1-E3, substantiation columns, accountant PDF

Section 8 UX Enhancements (`plan-task`)

#	Issue	Covers
8	[Dark Mode with Persistent Theme Toggle](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/8)	FR-UI2, CSS custom properties, WCAG AA
9	[Mobile-Responsive Layout (360px-Desktop)](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/9)	FR-UI3, hamburger nav, reflowing tables
10	[Polished UI with Shared Design Tokens](https://github.com/raghu-cmyk/mileage-tracker-v3/issues/10)	FR-UI1, cohesive tokens across all pages

Suggested Implementation Order

1. **#1** Design system (unblocks all UI work)
2. **#2** Authentication (foundation)
3. **#5** Reference data (unblocks trips & deductions)
4. **#3** Vehicles → **#4** Trips
5. **#6** Deductions → **#7** Export
6. **#10** Polished UI → **#8** Dark mode → **#9** Mobile responsive

...

mc done --summary "Created 10 planning issues (incl. design) for Build Mileage Tracker v3 with polished UI, dark mode, and mobile responsive layout"

Job 20260608T195548Z_30052b (implement)

Implemented **story_d3614a** – the full Mileage Tracker v3 MVP on `feat/mvp-v3` in `raghu-cmyk/mileage-tracker-v3`.

Deliverables

Functional MVP (adapted from v2, verified with 19 passing tests):

- Authentication (Argon2id, sessions, rate limiting)
- Vehicle management with odometer readings
- Trip management with §274(d) substantiation and audit trail
- Time-effective IRS mileage rates (2025/2026, DB-seeded)
- Deduction calculation and year-end summary
- IRS-compliant CSV and PDF export

Section 8 enhancements (v3-specific):

- **Ledger design tokens** in `app/static/css/tokens.css` per issue #1
- **Dark mode** with persistent `localStorage` toggle and `prefers-color-scheme` default (`app/static/js/theme.js`)
- **Mobile-responsive layout**: hamburger nav below 768px, card-list table reflow, 44px touch targets

Artifacts

Item	Location
Workspace code	`var/workspaces/20260608T195548Z_30052b/`
Output manifest	`artifacts/output.json`
Branch	`feat/mvp-v3` @ commit `1839b5d`
Pull request	[PR #11](https://github.com/raghu-cmyk/mileage-tracker-v3/pull/11) → `mc_webhook_test`

Verification

- 19/19 unit tests pass
- App imports cleanly (`Mileage Tracker v3 3.0.0`)

- No `TODO`/`FIXME`/`XXX`/`HACK` markers in workspace (acceptance gate satisfied)
- `mc report` and `mc done` called for job `20260608T195548Z\_30052b`

Job 20260608T195839Z_3093cd (validate)

Validation passed

Validated **raghu-cmyk/mileage-tracker-v3** on branch **feat/mvp-v3** using an in-process FastAPI `TestClient` (no long-lived server).

What was checked

1. **Cloned** `feat/mvp-v3` into `mileage-tracker-v3`
2. **Installed** dependencies in a virtualenv (`requirements.txt` + `httpx`, `pytest`)
3. **In-process route checks** – 15 routes, all OK:
 - `/health` → 200
 - `/` → 303 (unauthenticated) then 200 (authenticated)
 - `/login`, `/register` → 200
 - Static assets (`tokens.css`, `app.css`, `theme.js`) → 200
 - Auth flow: register → dashboard, vehicles, trips, summary, API → 200
 - HTML pages include doctype and theme toggle markup
4. **Unit tests** – 19/19 passed (rates, deductions, exports)

Evidence

- `validation\_evidence.json` – full status codes and results
- `VALIDATION\_PASSED` – one-line summary
- No TODO/FIXME/PLACEHOLDER markers found in source files

The app starts cleanly, seeds DB reference data on startup, and serves all key pages without errors.

Job 20260608T195952Z_526553 (deliver)

Delivery complete

Branch confirmed

`feat/mvp-v3` exists on `raghu-cmyk/mileage-tracker-v3` with validated work:

- **Head:** `feat/mvp-v3` (`1839b5d`)
- **Base:** `mc\_webhook\_test` (`457eac0`)
- **Commit:** Full MVP + Section 8 (Ledger design tokens, dark mode, mobile-responsive layout)
- **Gate:** `gate\_pr\_open.sh mc\_webhook\_test` → 1 open PR

Pull request

<https://github.com/raghu-cmyk/mileage-tracker-v3/pull/11>

Field	Value
Title	feat: Mileage Tracker v3 MVP – polished UI, dark mode, mobile responsive
Head	feat/mvp-v3
Base	mc_webhook_test

The PR body was updated with:

- **Closes** #2, #3, #4, #5, #6, #7, #8, #9, #10
- **Implements** design system from #1
- Validation notes (19 unit tests, app starts, no placeholder markers)
- Follow-ups for manual smoke tests and the PDF build report under `docs/build-reports/`

Note

`docs/build-reports/` is not yet on the branch; that is called out as a follow-up in the PR.

Mission Control

`mc done` signaled for job `20260608T195952Z\_526553`.

Job 20260608T200108Z_73108e (report)

4. Validation Results

Validate job (20260608T195839Z_3093cd): PASSED
 Method: FastAPI TestClient (in-process)
 Branch: feat/mvp-v3 | Repo: raghu-cmyk/mileage-tracker-v3

Route checks: 15/15 OK
Unit tests: 19/19 passed

- Gate outcomes:
- plan: gate_min_issues.sh (≥3 plan-task issues) – PASSED (10 issues created)
 - implement: no_todo_markers gate – PASSED
 - validate: VALIDATION_PASSED file – PASSED
 - deliver: gate_pr_open.sh mc_webhook_test – PASSED (PR #11 open)

Method	Path	Status	OK
GET	/health	200	✓
GET	/	303	✓
GET	/login	200	✓
GET	/register	200	✓
GET	/static/css/tokens.css	200	✓
GET	/static/css/app.css	200	✓
GET	/static/js/theme.js	200	✓
POST	/register	303	✓
...	+ 7 more routes	—	✓

5. Total Build Time

Earliest job start: 2026-06-08 19:53:22 UTC (plan)
Latest completed job finish: 2026-06-08 20:00:52 UTC (deliver)
Total elapsed: 7m 30s (excludes in-progress report job)

6. Application Stats (feat/mvp-v3)

Total source files (py/html/css/js/md/txt, excl. venv): 36
Total lines of code (py/html/css/js): 3,784
Unit tests passing: 19/19
Commit: 1839b5d
Features delivered: auth, vehicles, trips, rates, deductions, export,
design-tokens, dark-mode, mobile-responsive